

TIMOTHY BROWN HIGGINS

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Climate Fellow in the Department of Environmental Sciences and School of Data Science at the University of Virginia with a strong focus on enhancing severe weather forecasting through machine learning and merging numerical weather and climate data with natural language processing. Desired future work includes the application of cutting-edge research to address pressing challenges in climate science and improving predictive capabilities.

EDUCATION

University of Miami

Coral Gables, FL

Bachelor of Science, Atmospheric Sciences and Meteorology, Mathematics (applied)

Aug. 2016 – May 2020

University of Colorado

Boulder, CO

PhD, Atmospheric and Oceanic Sciences

Advisor: Aneesh Subramanian, PhD

Aug. 2020 – July 2025

PUBLICATIONS

- Higgins, T. B.**, Chapman, W. E., Subramanian, A. C., & Delle Monache, L. (2026). Generating a 1000-member Ensemble of Integrated Water Vapor Transport Forecasts with Diffusion. *Artificial Intelligence for the Earth Systems*, e250075. <https://doi.org/10.1175/AIES-D-25-0075.1>
- Higgins, T.**, Mamalakis, A., & Agarwal, C. (2026). SYNOPTIC BENCH : evaluating vision-language models on generating weather forecast discussions of the future. *Environmental Data Science*, 5, e19. <https://doi.org/10.1017/eds.2026.10050>
- Davis, I. W., Subramanian, A., **Higgins, T. B.**, Sengupta, A., & Delle Monache, L. (2026). Physics-Based Versus AI Weather Prediction Models: A Comparative Performance Assessment of Atmospheric River Prediction. *Geophysical Research Letters*, 53(4), e2025GL117609. <https://doi.org/10.1029/2025GL117609>
- Higgins, T. B.**, Subramanian, A. C., Watson, P. A. G., & Sparrow, S. (2025). Changes to Atmospheric River Related Extremes Over the United States West Coast Under Anthropogenic Warming. *Geophysical Research Letters*, 52(5), e2024GL112237. <https://doi.org/10.1029/2024GL112237>
- Higgins, T. B.**, Subramanian, A. C., Chapman, W. E., Lavers, D. A., & Winters, A. C. (2024). Subseasonal Potential Predictability of Horizontal Water Vapor Transport and Precipitation Extremes in the North Pacific. *Weather and Forecasting*, 39(6), 833–846. <https://doi.org/10.1175/WAF-D-23-0170.1>
- Higgins, T. B.**, Subramanian, A. C., Graubner, A., Kapp-Schwoerer, L., Watson, P. A. G., Sparrow, S., et al. (2023). Using Deep Learning for an Analysis of Atmospheric Rivers in a High-

Resolution Large Ensemble Climate Data Set. *Journal of Advances in Modeling Earth Systems*, 15(4), e2022MS003495. <https://doi.org/10.1029/2022MS003495>

Zheng, M., Delle Monache, L., Cornuelle, B. D., Ralph, F. M., Tallapragada, V. S., Subramanian, A., Haase, J.S., Zhang, Z., Wu, X., Murphy, M.J., **Higgins, T.B.**, Dehaan, L. (2021). Improved Forecast Skill Through the Assimilation of Dropsonde Observations From the Atmospheric River Reconnaissance Program. *Journal of Geophysical Research: Atmospheres*, 126(21), e2021JD034967. <https://doi.org/10.1029/2021JD034967>

FELLOWSHIPS

ATOC University Fellowship Fund

George Aiken Fellowship Award

UVA Environmental Institute Climate Fellowship

TEACHING EXPERIENCE

Graduate Teaching Assistant

January 2023 – May 2023

University of Colorado – Boulder, Boulder, CO

Teaching the weather lab for undergraduate students to learn how to collect data in science experiments.

Grader

University of Colorado – Boulder, Boulder, CO

January 2022 – present

Grading homework and exams for graduate students taking Geophysical Fluid Dynamics and Chaos and Predictability.

SOARS mentor

University of Colorado – Boulder, Boulder, CO

June 2021 – August 2021

Mentored an undergraduate intern for an analysis of several extreme weather events

Undergraduate Student Athlete Tutor

August 2018 – December 2018

University of Miami, Coral Gables, FL

Tutored NCAA Division I football players for Introduction to Weather and Climate

WORKSHOPS AND SUMMER SCHOOLS

NCAR ASP Graduate Student Visitor Program

March 2024 – September 2024

Mentor: Will Chapman, Ph.D.

Boulder, CO

ECMWF AR Reconnaissance Workshop

June 2023

Reading, UK

CW3E FIRO Workshop

July 2022

La Jolla, CA

NCAR ASP S2S Summer School

July 2021 - August 2021, July 2022

Boulder, CO

Project – Using various machine learning approaches to predict 2-m temperatures at the S2S scale and finding sources of skill with layer wise-relevance propagation

NOAA AI in Environmental Sciences Workshop

September 2021

Boulder, CO

PRESENTATIONS

15th International Conference on Climate Informatics – April 2026

Lausanne, Switzerland

SynopticBench: Evaluating Vision-Language Models on Generating Weather Forecast
Discussions of the Future

**Machine Learning for Actionable Climate Science Gordon Research Conference Poster
Presentation – June 2025**

Smithfield, RI

Generating a 1000-member Ensemble of Integrated Water Vapor Transport Forecasts with
Diffusion

2024 Annual American Geophysical Union Meeting Oral Presentation – December 2024

Washington, D.C.

Diffusion Ensemble Generation of West-WRF Integrated Vapor Transport Forecasts

**Weeks 3-4/S2S Webinar by OAR/WPO S2S Program and NWS/OSTI-Modeling
Distinguished Speaker Oral Presentation – July 2024**

Boulder, CO

An analysis of differences in predictability of precipitation and IVT extremes in S2S forecasts
and their connection to the jet stream over the North Pacific

**CESM Climate Variability and Change Working Group Meeting Oral Presentation–
March 2024**

Boulder, CO

The impact of climate change on atmospheric river extremes from a unique large-ensemble
atmospheric model output

CESM Earth System Prediction Working Group Meeting Oral Presentation– March 2024

Boulder, CO

An analysis of differences in predictability of precipitation and IVT extremes in S2S forecasts
and their connection to the jet stream over the North Pacific

2023 Annual American Geophysical Union Meeting Poster Presentation – December 2023

San Francisco, CA

The impact of climate change on atmospheric river extremes from a unique large-ensemble atmospheric model output

World Climate Research Program Open Science Conference Poster Presentation– October 2023

Kigali, Rwanda

The impact of climate change on atmospheric river extremes from a unique large-ensemble atmospheric model output

University of Reading S2S Summit Poster Presentation – June 2023

Reading, UK

An analysis of differences in predictability of precipitation and IVT extremes in S2S forecasts and their connection to the jet stream over the North Pacific

ECMWF Atmospheric Rivers Reconnaissance Workshop Oral Presentation – June 2023

Reading, UK

An analysis of differences in predictability of precipitation and IVT extremes in S2S forecasts and their connection to the jet stream over the North Pacific

103rd Annual American Meteorological Society Meeting Poster Presentation – January 2023

Denver, CO

An analysis of differences in predictability of precipitation and IVT extremes in S2S forecasts and their connection to the jet stream over the North Pacific

ESSS Poster Conference Poster Presentation – December 2022

Boulder, CO

An analysis of differences in predictability of precipitation and IVT extremes in S2S forecasts and their connection to the jet stream over the North Pacific

International Atmospheric Rivers Conference 2022 Oral Presentation – October 2022

Santiago, Chile

The impact of climate change on atmospheric river extremes from a unique large-ensemble atmospheric model output

CW3E Annual Meeting Poster Presentation – June 2022

La Jolla, CA

An analysis of differences in predictability of precipitation and IVT extremes in S2S forecasts and their connection to the MJO and PNA.

European Geophysical Union General Assembly 2022 Oral Presentation – May 2022

Vienna, Austria

Discussion of the speed and effectiveness of using deep learning to track atmospheric rivers in a large ensemble high-resolution climate dataset

University of Colorado – Boulder Hydrologic Sciences Symposium Oral Presentation – April 2022

Boulder, CO

Discussion of the speed and effectiveness of using deep learning to track atmospheric rivers in a large ensemble high-resolution climate dataset

NOAA AI in Environmental Sciences Workshop Oral Presentation – September 2021

Boulder, CO

Discussion of the speed and effectiveness of using deep learning to track atmospheric rivers in a large ensemble high-resolution climate dataset

19th Annual AMS Student Conference Poster Presentation – January 2020

Boston, MA

Case study of the effect of assimilated dropsonde data on the accuracy of WRF model output

FIRO Conference Poster Presentation – August 2019

La Jolla, CA

Analysis of the effect of assimilated dropsonde data on the accuracy of WRF model output for an Atmospheric River event that occurred on February 3rd, 2018

COMPUTING SKILLS AND EXPERIENCE

Programming languages:

MATLAB, Python

Technical skills:

Linux (HPC, bash scripting), Git, Parallel computing

Machine learning skills:

Diffusion, Convolutional Neural Networks, Self-organizing maps, Vision Language Models

Tools and libraries:

PyTorch, TensorFlow, SciPy, NumPy, Pandas, xarray

Data types:

NetCDF, GRIB2

Models worked with:

CESM2, HadAM4, ERA5, ECMWF, West-WRF, MERRA-2

Models run:

CESM2

Soft skills:

Teaching, communication, collaboration, creativity, problem-solving

